Jia-Wei Liao

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Summary

Jia-Wei Liao is a Ph.D. candidate in Computer Science at National Taiwan University and a Research Assistant at Academia Sinica. His research focuses on generative models, particularly diffusion models and flow matching, with applications in controllable generation, inverse problems, trustworthy AI, and robot manipulation. His work has been published at leading international conferences, including CVPR, ECCV, WACV, AAAI, and CoRL. He received his M.S. degree in Applied Mathematics from National Yang Ming Chiao Tung University, where he worked on computational geometry and medical image analysis. His background in applied mathematics and numerical computation continues to shape his research on structured and geometry-aware generative modeling.

Research Interests

- **Generative Modeling:** Developing diffusion and flow matching models, particularly through inference-time guidance and structured prior modeling, to improve generation quality, controllability, and efficiency.
- **Visual Perception:** Recovering high-quality images from degraded, incomplete, or noisy observations, and enabling machines to build computational representations of the world's structure and state.
- **Trustworthy AI:** Investigating the safety of generative models, including adversarial attacks and machine unlearning, to uncover their vulnerabilities and develop more reliable models and algorithms.

Education

National Taiwan University (NTU) *Sep. 2022 - Present*
Ph.D. Candidate in Computer Science and Information Engineering | CMLab GPA: 4.29 / 4.30 (46 Credits)

National Yang Ming Chiao Tung University (NYCU) *Sep. 2020 - Aug. 2022*
M.S. in Applied Mathematics | GIMI LAB GPA: 4.29 / 4.30 (36 Credits)

National Taiwan Normal University (NTNU) *Aug. 2016 - Jun. 2020*
B.S. in Mathematics | Educational Program | Scientific Computation Program GPA: 4.06 / 4.30 (154 Credits)

Work Experience

Academia Sinica *Jul. 2023 - Present*
Research Assistant Taipei, Taiwan
Developed generative model algorithms, including an aesthetic QR code generator, inference-time controllable generation techniques, and protective AI algorithms, with research outcomes published in top-tier AI/CV conferences

SinoPac Holdings AI Team *Dec. 2024 - Mar. 2026*
Research Intern Taipei, Taiwan
Developed tabular foundation models, encompassing data preprocessing, model architecture design, loss function formulation, and training strategies, with results published in ML workshops

Intel Wireless Solution Group *Jul. 2024 - Aug. 2024*
Software Engineer Intern Taipei, Taiwan
Designed a Retrieval-Augmented Generation (RAG) system based on the Llama 3.1 model for an internal database, and built a full-stack application

Appier AiDeal Team *Aug. 2022 - Jun. 2023*
Data Analyst Intern Taipei, Taiwan
Developed a predictive model to identify user purchase tendencies, defined key evaluation metrics, and implemented feature engineering and selection methods, achieving a 3% improvement in key business metrics

National Center for Theoretical Sciences (NCTS) *Jul. 2020 - Aug. 2020*
Student Researcher Taipei, Taiwan
Developed a variational approach for image denoising and an image stitching algorithm

Selected Publications (* indicates equal contribution)

[1] **Jia-Wei Liao**, Li-Xuan Peng, Mei-Heng Yueh, Ming Sun, Cheng-Fu Chou, Jun-Cheng Chen. DiffRGD: An Inference-time Diffusion Guidance Through Riemannian Gradient Descent. *ECCV 2026*.

[2] Chien-Sheng Chiang*, Kuan-Hsun Tu*, **Jia-Wei Liao***, Cheng-Fu Chou, Tsung-Wei Ke. COT-FM: Cluster-wise Optimal Transport Flow Matching. *CVPR 2026*.

[3] Ju-Hsuan Weng*, **Jia-Wei Liao***, Cheng-Fu Chou, Jun-Cheng Chen. M-ErasureBench: A Comprehensive Multimodal Evaluation Benchmark for Concept Erasure in Diffusion Models. *WACV 2026*.

[4] Chun-Yen Shih*, Li-Xuan Peng*, **Jia-Wei Liao**, Ernie Chu, Cheng-Fu Chou, and Jun-Cheng Chen. Pixel Is Not A Barrier: An Effective Evasion Attack for Pixel-Domain Diffusion Models. *AAAI 2025*.

[5] **Jia-Wei Liao**, Winston Wang*, Tzu-Sian Wang*, Li-Xuan Peng*, Ju-Hsuan Weng, Cheng-Fu Chou, and Jun-Cheng Chen. DiffQRCoder: Diffusion-based Aesthetic QR Code Generation with Scanning Robustness Guided Iterative Refinement. *WACV 2025*.

[6] Jung-Hao Kao*, Jing-En Huang*, **Jia-Wei Liao***, Tzuhsuan Huang, Yu-Lun Liu, Cheng-Fu Chou, Jun-Cheng Chen. QVR3D: Query-Guided Visual Reasoning for Training-Free 3D Grounding. *WACV 2027*.

[7] Huang-Yu Chen, Jia-Fong Yeh, **Jia-Wei Liao**, Pin-Hsuan Peng, Winston H Hsu. Distribution Discrepancy and Feature Heterogeneity for Active 3D Object Detection. *CoRL 2024*.

[8] **Jia-Wei Liao**, Tsung-Ming Huang, Tiexiang Li, Wen-Wei Lin, Han Wang, Shing-Tung Yau. An UNet-based Brain Tumor Segmentation Framework via Optimal Mass Transportation Pre-processing. *MICCAI BrainLes 2022 Workshop*.

Honors and Awards

- TWSIAM Thesis Award; Most Popular Award *Jun. 2025*
- AI CUP Multimodal Pathological Voice Classification Competition, Gold Medal *Mar. 2024*
- AI CUP Agricultural Crop Status Survey Image Recognition Competition, Honorable Mention Award *Mar. 2023*
- AI CUP Smart Counting of Unmanned Aerial Vehicles Competition, Honorable Mention Award *Mar. 2023*
- AI CUP Explanatory Data Labeling for Natural Language Understanding, Honorable Mention Award *Mar. 2023*
- AI CUP STAS Competition Merit Award and MATLAB Outstanding Application, First Prize *Mar. 2023*

Teaching Assistant Experience

- **CSIE5441: Diffusion Models and Their Applications** Shih-Hsin Wang
  NTU Computer Science - Fall 2026
- **CSIE5421: Embodied Vision** Tsung-Wei Ke
  NTU Computer Science - Spring 2026, Spring 2025
- **LAI11380: Robotic Navigation and Exploration** Chun-Yi Lee
  NTU Computer Science - Spring 2026, Spring 2025
- **CSIE5117: Robot Perception and Learning** Tsung-Wei Ke
  NTU Computer Science - Fall 2026, Fall 2025, Fall 2024
- **CSIE5043: Machine Learning** Chun-Yi Lee
  NTU Computer Science - Fall 2025
- **CSIE5126: Digit, AI, and Quantum Computing** Bor-Sung Liang
  NTU Computer Science - Fall 2023, Fall 2022
- **CSIE2120: Linear Algebra** Wen-Chin Chen
  NTU Computer Science - Fall 2022

Selected Talks and Presentations

- Exploring Generative Diffusion Models for Personalized Aesthetic QR Code Design, *PyCon 2025*.
- Training-Free Scanning Robustness Guided Diffusion Model for Aesthetic QR Code Generation, *sciwork 2024*.
- Multimodal Pathological Voice Classification, *2024 AI CUP Awards Ceremony*.
- Taiwan-LLM Tutor: Large Language Models for Taiwanese Secondary Education, *SITCON 2024*.
- How can we perfectly and rapidly stitch images? Exploring improved end-to-end techniques, *sciwork 2023*.
- Geometric Transformations in the Digital World, *SITCON 2022*.

Academic Service

- **Conference Reviewer:** NeurIPS, ICLR, ICML, CVPR, ECCV, WACV, ICIP
- **Journal Reviewer:** IEEE T-IFS